

adjusting heuristic causes us to stick with our first impression, even in the face of additional evidence that should cause us to change our view.

Each heuristic manifests in a variety of different ways. Most of the time they are very useful and efficient but, in assessing probabilities and predicting uncertain future states, they can lead us into systematic and predictable errors. The problem, writes Leonard Mlodinow in his wonderful book *The Drunkard's Walk*, is that the mechanisms by which we analyze uncertain, probabilistic conditions are an “intricate product of evolutionary factors, brain structure, personal experience, knowledge and emotion” so complex that “different structures within the brain come to different conclusions.”⁴³ The confusion is often that the logical left hemisphere of our brains looks for a pattern, while the right hemisphere behaves more intuitively. Mlodinow gives the following example of the tension between our pattern-seeking behavior and our more intuitive impulses: The game is called *probability guessing*. Subjects are shown lights that have two colors, say red and green. The lights flash so that one color appears more frequently than the other, but without any pattern. So, for example, red might appear twice as often as green in a sequence like red-red-green-green-red-red-green-red-red-red-green and so on. After watching for a period of time, the subject must guess which color will appear next. Mlodinow tells us that there are two basic strategies. The first is to always guess the color that appears most frequently. This is the strategy favored by rats and other animals, and our intuitive hemisphere. If we employ this strategy, we are guaranteeing a certain level of success, but can do no better. If red appears twice as often as green, two-thirds of our guesses will be correct, and one-third will be incorrect. The other strategy is to figure out the pattern. This is the strategy favored by MBA students and our logical hemisphere. If the lights flash in a pattern, and we can figure out that pattern, we can guess correctly every single time. If the lights flash randomly we can guess correctly only to the extent that we favor the color that flashes more regularly. We make an intuitive tradeoff when we try to guess correctly—we forgo the guaranteed, but known, error rate for the possibility that we get all correct. This is the reason that rats tend to outperform MBA students in these types of experiments. Our native intuition is wrong, and our reasoning apparatus poor under conditions of uncertainty.

There are few arenas that allow us to demonstrate the full range of our heuristic pathologies quite like the stock market. The research is clear that value stocks—defined as a low price in relation to some fundamental measure like earnings, book value, or cash flow—outperform glamour stocks, which have a high price in relation to those fundamental measures. Yet we are intuitively attracted to glamour stocks. We like them because they have done well in the recent past, either in terms of stock price performance